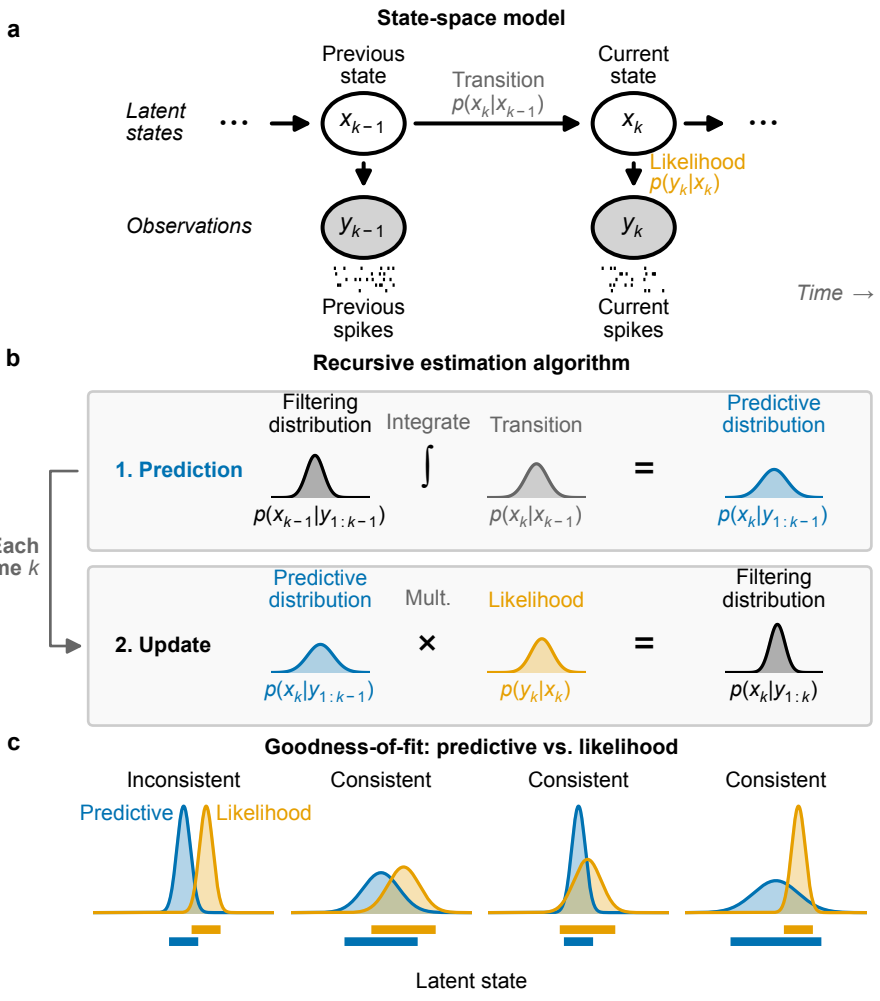

a
State-space model
Previous state
Transition $p(x_k|x_{k-1})$
Current state
Latent states
x_{k-1}
x_k
Likelihood $p(y_k|x_k)$
Observations
y_{k-1}
y_k
Previous spikes
Current spikes
Time →
b
Recursive estimation algorithm
Filtering distribution
Integrate
Transition
Predictive distribution
1. Prediction
$p(x_{k-1}|y_{1:k-1})$
$p(x_k|x_{k-1})$
$p(x_k|y_{1:k-1})$
Each time k
Predictive distribution
Mult.
Likelihood
Filtering distribution
2. Update
$p(x_k|y_{1:k-1})$
$p(y_k|x_k)$
$p(x_k|y_{1:k})$
c
Goodness-of-fit: predictive vs. likelihood
Inconsistent
Consistent
Consistent
Consistent
Predictive
Likelihood
Latent state